

MUHAMMAD HILMAN BIN MOHD AMIN

Electronic Engineering Technology Student | Embedded Systems & Computer Vision
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SUMMARY

Motivated Electronic Engineering Technology (Communication and Computer) student currently in the final semester at Universiti Tun Hussein Onn Malaysia. Experienced in academic and technical projects involving communication systems, embedded systems, and computer networking. Skilled in circuit design, microcontroller programming, and basic networking tasks. Preparing to undergo industrial training in the upcoming semester and seeking an internship opportunity to gain practical experience and further develop my technical skills.

EDUCATION

Universiti Tun Hussein Onn Malaysia (UTHM) <i>Bachelor of Electronic Engineering Technology (Communication and Computer) — CGPA 3.38/4.00</i>	<i>Oct 2022 – Present</i>
Polytechnic Sultan Mizan Zainal Abidin <i>Diploma in Information Technology — CGPA 3.38/4.00</i>	<i>June 2018 – May 2022</i>

CERTIFICATES

• STM32 Programming Certified (Beginner)	June 2024
• Introduction to AI, Google	Oct 2025

TECHNICAL SKILLS

GUI & Software	Python, PyQt5, C++, Visual Studio Code
Automation & CV	OpenCV, YOLOv8, Manufacturing Analytics (OEE, UPH, GSPH)
Embedded Systems	Arduino IDE, ESP32, STM32, Circuit Design, CST Studio Suite, Digital Electronics
Database	MySQL, phpMyAdmin
Productivity	Microsoft Word (Excellent), Excel, PowerPoint
Languages	Malay (Native), English (Proficient), German (Basic)

PROJECT EXPERIENCE

Smart Factory System <i>Project Member Peps-Jv (M) Sdn. Bhd. & UTHM — Selangor, Malaysia</i>	<i>Feb 2026 – Present</i>
- Designing and developing a Smart Factory dashboard using PyQt5 to monitor real-time manufacturing operations on an active production line. - Integrating computer vision (OpenCV/YOLOv8) to automate object counting and tracking from a live camera feed, reducing reliance on manual line counts. - Building automated calculation modules for key manufacturing metrics, including Units Per Hour (UPH) and Overall Equipment Effectiveness (OEE). - Implementing data tracking for production records, rejection logs, and downtime durations to support operational decision-making.	
RFID-Based Library Management System for Automated Book Tracking and Enhanced Security <i>Final Year Thesis — Johor, Malaysia</i>	<i>March 2025 – Present</i>
- Developing an RFID system for real-time, automated tracking of books in the library. - Building a centralized MySQL database to manage book and user records efficiently. - Implementing RFID-based authentication for secure, seamless check-in/check-out.	

Microcontroller and Microprocessor: Line Following Mobile Robot with Metal Detection

January 2024

Project Member | UTHM — Johor, Malaysia

- Built an autonomous mobile robot on an embedded system platform, integrating sensors for line tracking and metal object detection.
- Programmed motor control and sensor logic in Arduino, improving path-following accuracy and detection reliability through iterative testing.

WORKING EXPERIENCE

Sales Assistant, Ninja Code HQ

June 2022 – August 2022

- Managed daily retail operations and handled customer inquiries, driving walk-in sales.
- Processed POS transactions (cash/cashless) and prepared daily sales summaries.

Sales Assistant, BSSE Computer Centre

August 2021 – December 2021

- Advised customers on hardware/software selection to support informed purchasing decisions.
- Managed POS transactions and inventory, maintaining organized stock control.

REFERENCE

Dr. Mohd Shamian Bin Zainal

Senior Lecturer/ Academic Advisor

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